Sheet 4

Conjugation, ring currents and phase choices

1. Let C be conjugation in the σ_z basis. Compute $C\sigma_j C^{-1}$ and its action on a Bloch vector. Check directly that $C(i\psi) = -iC\psi$ and that ray transition probabilities are preserved.
2. On the ring take $\theta = \pi/3$. Compute the three levels for $k = -1, 0, 1$ in units $\hbar^2\pi^2/(2mL^2)$ and the ground current with $\theta = -q\Phi/\hbar$. Identify their time-reversed partners. At $\theta = \pi$, give the two ground-branch currents and explain why no unique derivative of the ground energy specifies the current there.
3. If U_t is a genuine unitary group, define $\mathcal{V}_t = e^{i\alpha t^3}U_t$, where α is real and has units time^{-3}. Compute its multiplier and give a rephasing restoring the group law. Does strong continuity of this family alone make it a group?